

SM87BG & SM87PB manual call point range

Ex db, Ex ia, weatherproof



Overview

These manual fire alarm, emergency shutdown break glass and push button units have been designed for the most arduous environmental conditions.

The units are both easy to install and maintain. Intrinsically safe Ex ia and flameproof Ex db versions of each model are available.

A choice of either stainless steel or alloy makes the range suitable for either the offshore or onshore industries. Stainless steel, one of the most durable materials available on the market, is both hard wearing and corrosion resistant, increasing the life of products in harsh environments and therefore reducing maintenance costs.

Features

- Zone 0, Zone 1 and Zone 2 use*
- ATEX certified
- UL listed for USA and Canada (SM87PBL only). Class I, Div 1, Groups C & D
- ULC certified (SM87PBL only) for Class I, Zone 1 Groups C & D
- CSA certified
- IECEx certified
- Brazilian (INMETRO) certified
- PESO certified (Ex db only)
- SIL 2 certified
- IP66 and IP67
- DNV type approval (IEC 60945) (SM87PB only)
- Certified temperature: -55°C to +70°C*
- Stainless steel or marine grade alloy
- Robust yet lightweight
- Easy to maintain

*Model dependent



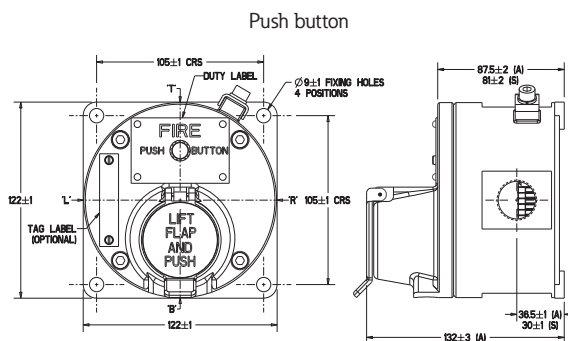
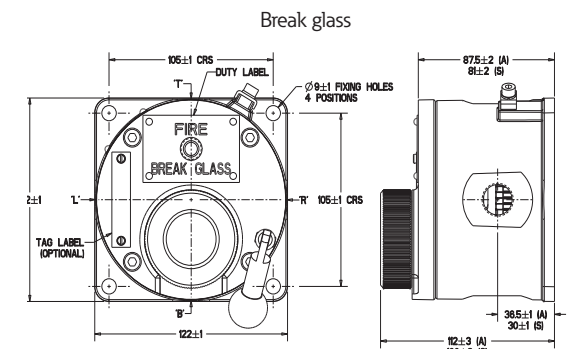
Certifications

ATEX Ex db	Cert. no. Baseefa03ATEX0075. ATEX Approved Ex II 2GD Certified to: EN60079-0, EN60079-1, EN60079-31 Ex db IIC T5/T6 Gb, Ex tb IIIC T85°C/T100°C Db. IP66/IP67
ATEX Ex ia	Cert no. Baseefa 02ATEX0152X. ATEX Approved Ex II 1GD Certified to: EN60079-0, EN60079-11 Ex ia IIC T4 Ga, Ex ia IIIC T135°C Da
IECEx Ex db	Cert. no. IECEx BAS 09.0060 Certified to: EN60079-0, EN60079-1, EN60079-31 Ex db IIC T5/T6 Gb, Ex tb IIIC T85°C/T100°C Db. IP66/IP67
IECEx Ex ia	Cert. no. IECEx BAS 10.0033X Certified to: IEC60079-0, IEC60079-11, Ex ia IIC T4 Ga, Ex ia IIIC T135°C Da
UL	Listing no. E186629 UL listed to Class 1, Div 1. Groups C & D. (SM87PBL)
ULC	Cert. no. 20091023-E320282 ULC certified for Class I, Zone 1 Groups C & D (SM87PBL)
CSA Explosion proof	Class 1, Div 1 & 2. Group D Ex d IIC T5/T6 Gb
CSA Intrinsically safe	Class 1, Div 1 & 2. Groups A-D
INMETRO Ex d	Ex db IIC T5/T6 Gb
Type Approvals	DNV (IEC 60945) (Stainless Steel SM87PB only)
SIL	SIL2 Certification to IEC61508
CCC	Ex db IIC T5/T6 Gb, Ex tb T85°C/T100°C Db

Specifications

Material	Grade 316 ANC4B stainless steel or LM 25TF marine grade alloy
Finish	Paint finish as standard or to customer specification
Voltage	Ex db 24V a.c./d.c. Ex ia 28V
Rating	2A
Switches	2 pole c/o, wired to terminals Optional up to 4 pole (UL version 2 pole only)
Optional indicator	A red high intensity LED can be fitted for alarm indication As standard the LED is not provided with over current protection. The forward current (If) should be limited to 20mA
Certified temp	Ex db* -55°C to +70°C. Ex ia* -55°C to +60°C UL -20°C to +55°C (LED version only) -40°C to +70°C, -20°C to +55°C (LED version only) CSA Explosion proof - Class 1, Div 1 & 2. -50°C to +55°C CSA - Ex d IIC -55°C to +70°C CSA Intrinsically safe - Class 1, Div 1 & 2 -50°C to +40°C *Note: includes ATEX, IECEx & INMETRO versions
Weight	3.8 kg. steel (approx.) or 2.5 kg. alloy (approx)
Ingress protection	IP66 & IP67. SM87PB IP68 (35m for 40 hours)
Entries	Up to 4 x M20 or M25 ISO Ex d/Ex ia Up to 4 x ½" or ¾" NPT UL Stainless steel version comes with two entries as standard. Units ordered with a single entry ordering code are supplied with 2x entries inc. plug on the opposite side
Terminals	Will accept up to 1.5mm² cable
Resistor values	470R minimum (d.c. & I.S. units only)

General arrangement drawing (all dimensions in mm)



Both the Ex ia units and the Ex db units have the same external appearance. Also the internal components are identical throughout the range. Each unit can be wired for either NO, NC or CO contacts to customer specification

Ordering requirements

The following code is designed to help in selection of the correct unit. Build up the reference number by inserting the code for each component into the appropriate box

Unit	Model	Material	Certification	Entries	Duty label	Tag label	Options	Finish
SM87								
		Material* Stainless steel S Alloy A	Code S A	*UL version only available in alloy				
Model* Break glass latching BGL Lift flap break glass latching LBGL Push button latching key reset PBL Push button only momentary action PBM		Certification Ex ia I Ex d D IECEx - Ex ia IJ IECEx - Ex db DJ UL UL ULC UC CSA - Intrinsically Safe IC CSA - Ex proof DC INMETRO - Ex db DM CCC DQ		Code I D IJ DJ UL UC IC DC DM DQ	Entries M20 1 M25 2 ½" NPT 3 ¾" NPT 4 Other 5			
				Position Top *T Bottom *B RHS *R LHS *L	Duty label Fire breakglass 1 Push button 2 Fire push button 3 ESDO abandon 4 Burning house logo 5 Fire/FEU 6† Other O* None N			
				*Please specify †This option must be used when ordering a red callpoint for ULC Fire Alarm applications				
				Tag label Required Y* Not required N				
				*Please specify				
				Options Not required N LED A Resistor series G* Resistor EOL H* Diode D* 3 pole c/o switch T 4 pole c/o switch F				
				*Please specify				
				Finish Red R Blue B Yellow Y Yellow/black stripes X Special S*				
				*Please specify				

*UL and ULC versions only available in PBL model

*Prefix position with size code i.e. 1T1B = M20 Top and Bottom